

1. In order to address the data-shared problem, we can use **Mutual exclusion**
2. The question is about the interleaved execution of the two threads (Payroll and ATM withdrawal) presented in the lecture. If the initial balance B is \$120, what will be the final value of B after the interleaved execution of both threads.

B = \$120

•Interleaved Execution

•Load Rg, B

•Sub Rg, 60

Interrupt

•Load Rg, B

•Add Rg, 1000

•Store B, Rg

Return from int..

•Store B, Rg

60

3. When _____ threads of execution access _____ variables, the integrity of these variables may be _____ if the access to data is not _____.
cooperating
shared
violated
coordinated
4. The shared-data problem or (race condition) arises when:
Multiple threads execute concurrently
Threads access and manipulate shared data
5. A critical region is also called a _____. (two words)
critical section
6. Mutual exclusion means _____.
A thread will not manipulate some shared data D while another is also manipulating the same data D.
7. If a solution to Mutual Exclusion meets the progress requirement, this means that
a thread NOT in its critical region cannot block another thread from entering its respective critical region
8. A solution to mutual exclusion must have the following components: initialization, entry code, and _____ code.
exit
9. A critical region is _____

a piece of code manipulating data shared by at least two threads.

10. A solution to mutual exclusion must have the following components: initialization, _____ code, and exit code.

entry

11. A solution to mutual exclusion must have the following components:

Initialization

Exit Code

entry code

12. A solution to the mutual exclusion must meet the following requirements:

Progress

CPU-speed independent

Mutual Exclusion